CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 15: Similarity (With Applications to Maps and Models)

Previous year board questions, topic-wise: 13 questions

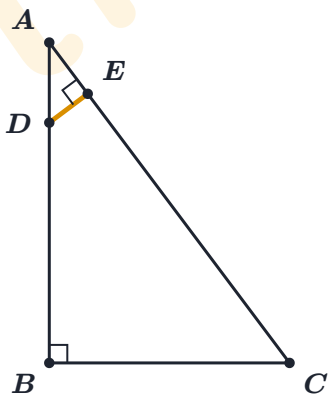
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Similar triangles

1. $\triangle ABC \sim \triangle EFG$ with $\angle ABC = \angle EFG = 60°$, $AB = 15$ cm, $BC = 3$ cm and $EF = 75$ cm. Find FG. (a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm ICSE 2025, 1 mark

2 Maps and models

2. In $\triangle ABC$, $\angle ABC = 90°$, $AB = 20$ cm and $AC = 25$ cm. D is a point on AB and E is a point on AC with $DE \perp AC$ and $DE = 3$ cm. (a) Prove that $\triangle ABC \sim \triangle AED$. (b) Find BC, AD and AE. (c) $BCED$ is a plot of land drawn on a map, and its actual area is 576 m². Find the scale factor of the map. ICSE 2025, 4 marks



3. A model of a high-rise building is made to a scale of $1 : 50$. (a) The model is 0.8 m high; find the height of the building. (b) A flat in the building has a floor area of 20 m²; find its floor area in the model. ICSE 2020, 3 marks

4. A model of a building is made with scale factor $1 : 30$. (a) The model is 80 cm high; find the height of the building in metres. (b) A tank on top of the building has a volume of 27 m³; find the volume of the tank on the model. ICSE 2019, 3 marks

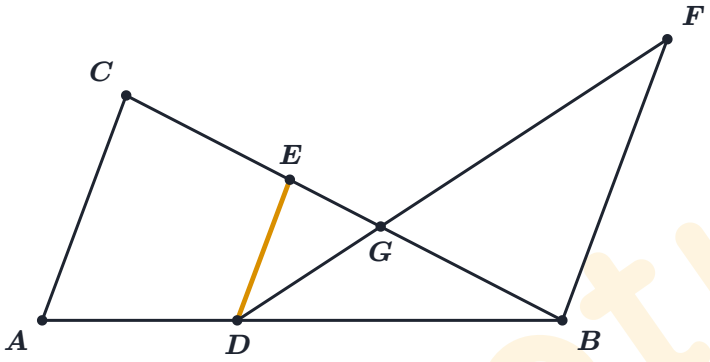
5. On a map drawn to a scale of $1 : 50{,}000$, a rectangular plot $ABCD$ has $AB = 6$ cm and $BC = 8$ cm. Find (a) the actual length of the diagonal AC in km (b) the actual area of the plot in km². ICSE 2018, 3 marks

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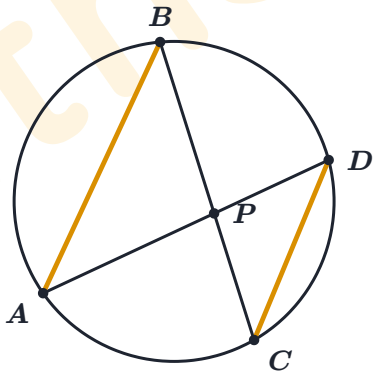
3 Conditions for similarity

6. $\triangle ABC \sim \triangle PQR$. AD bisects $\angle BAC$ and meets BC at D; PS bisects $\angle QPR$ and meets QR at S. Which statement is true? (a) $\triangle ABC \sim \triangle PQS$ (b) $\triangle ABD \sim \triangle PQS$ (c) $\triangle ABD \sim \triangle PSR$ (d) $\triangle ABC \sim \triangle PSR$ ICSE 2024, 1 mark

7. AC, DE and BF are parallel, with D on AB and E on CB. DF meets CB at G. Given $AC = 24$ cm, $EG = 8$ cm, $GB = 16$ cm and $BF = 30$ cm, (a) prove that $\triangle GED \sim \triangle GBF$ (b) find DE (c) find $DB : AB$. ICSE 2023, 4 marks



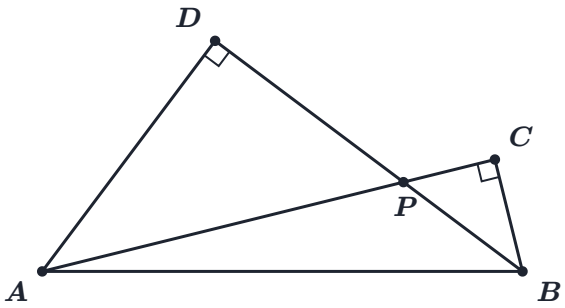
8. Chords AD and BC of a circle intersect at P. $AB = 9$ cm, $PA = 7.5$ cm and $PC = 5$ cm. (a) Prove that $\triangle PAB \sim \triangle PCD$. (b) Find CD. (c) Find area of $\triangle PAB$: area of $\triangle PCD$. ICSE 2020, 3 marks



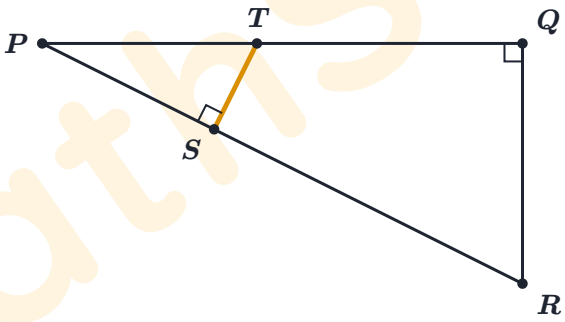
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4 Areas of similar triangles

9. $\triangle ADB$ and $\triangle ACB$ are right-angled at D and C, on the same side of AB, and AC meets BD at P. $AB = 10$ cm, $AD = 6$ cm, $BC = 2.4$ cm and $DP = 4.5$ cm. (a) Prove that $\triangle APD \sim \triangle BPC$. (b) Find BD and PB. (c) Hence find PA. (d) Find area of $\triangle APD$: area of $\triangle BPC$. ICSE 2024, 4 marks

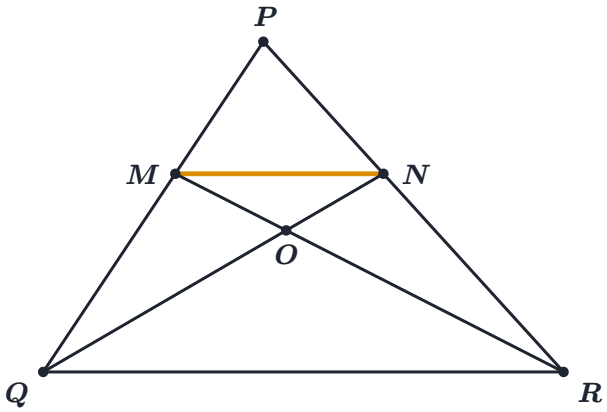


10. In $\triangle PQR$, $\angle PQR = 90°$. T is a point on PQ and S is a point on PR with $\angle PST = 90°$. $PQ = 5$ cm and $PS = 2$ cm. (a) Prove that $\triangle PQR \sim \triangle PST$. (b) Find area of $\triangle PQR$: area of quadrilateral $SRQT$. ICSE 2019, 3 marks



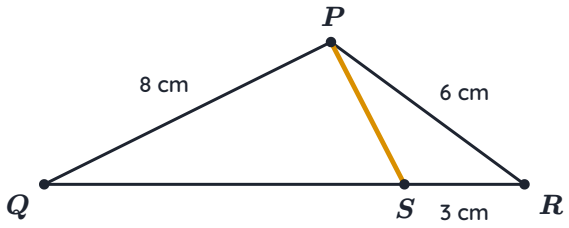
11. In $\triangle PQR$, M lies on PQ and N lies on PR with $MN \parallel QR$ and $\frac{PM}{MQ} = \frac{2}{3}$. MR and NQ meet at O. (a) Find $\frac{MN}{QR}$. (b) Prove that $\triangle OMN \sim \triangle ORQ$. (c) Find area of $\triangle OMN$: area of $\triangle ORQ$.

ICSE 2018, 3 marks



12. S is a point on the side QR of $\triangle PQR$ with $\angle PSR = \angle QPR$. $QP = 8$ cm, $PR = 6$ cm and $SR = 3$ cm. (a) Prove that $\triangle PQR \sim \triangle SPR$. (b) Find QR and PS. (c) Find $\dfrac{\text{area of } \triangle PQR}{\text{area of } \triangle SPR}$.

ICSE 2017, 3 marks



5 Basic proportionality theorem

13. In $\triangle PAB$, C is a point on PA and D is a point on PB with $\angle BAP = \angle DCP = 70°$. If $PC = 6$ cm and $CA = 4$ cm, find $PD : DB$. (a) $5 : 3$ (b) $3 : 5$ (c) $3 : 2$ (d) $2 : 3$ ICSE 2023, 1 mark

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